

Assignment 4: Generalized DiD with Cohort Exposure – Indonesia School Construction

Advanced Methods for Data Analysis in Public Policy Research FS 26 — ETH Zürich

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Application Background

This assignment uses the application context of Duflo (2001), who evaluates a large school construction program in Indonesia and its effects on education and adult earnings. The identification strategy relies on two sources of variation: (i) districts received different program intensity (more/less school construction), and (ii) only younger birth cohorts were exposed while older cohorts serve as a control group. This is a generalized Difference-in-Differences (DiD) design with cohort exposure and continuous treatment intensity.

Research Questions:

1. Did school construction increase years of schooling for exposed cohorts?
2. Did school construction raise adult wages (log wages) for exposed cohorts?

Instructions

- Provide all code and answers within the provided markdown template.
- Ensure that you formulate a written answer where asked. Be concise.
- **Pass requirements:**
 - The analysis must be replicable on any machine. Use relative file paths when loading the data.
 - Completion of all questions is mandatory for a passing grade.
- **When you are done:** Submit the Markdown script **and** the compiled HTML file on Moodle.
- **Submission deadline:** 04.05.2026, 9:00 am.
- **Presentation:** The assignment will be discussed during the exercise session on 11.05.2026. Be prepared to present the assignment in class. Every group member must be able to discuss and defend any part of the assignment.

Hint: Because `school_intensity` varies at the district level, cluster standard errors by district throughout.

Exercises

1. Descriptives and intuition

- a) Summarize the distribution of `school_intensity` across districts (mean, sd, min, max). Are there districts with zero intensity?
- b) Compare mean `years_schooling` for exposed vs non-exposed cohorts. Do the same for `ln_wage`.

- c) Explain why older cohorts can serve as a control group and what variation identifies the program effect.

2. Regression setup (generalized DiD)

- a) Estimate the baseline regression for schooling:

$$years_schooling_{dc} = \alpha_d + \gamma_c + \beta(exposed_c \times school_intensity_d) + \varepsilon_{dc},$$

where α_d are district fixed effects and γ_c are cohort fixed effects.

- b) Explain the interpretation of β .
- c) State the identifying assumption in this context (a ‘parallel trends’ style assumption).

3. Baseline estimates: schooling

- a) Estimate the model in (2). Use district and cohort fixed effects.
- b) Report $\hat{\beta}$ and interpret it as the effect of a one-unit increase in intensity for exposed cohorts.
- c) Translate the estimate into the effect of 1 additional school per 1,000 children (same unit as intensity).

4. Adding controls and robustness

- a) Add *baseline_dev* and *urban_share* to the regression (still with district and cohort fixed effects).
- b) Does $\hat{\beta}$ change meaningfully? Discuss.
- c) Do these controls change the identifying assumption?

5. Labor market outcome: wages

- a) Repeat the analysis for *ln_wage* as the outcome using the same specification.
- b) Report $\hat{\beta}_{wage}$ and interpret it.
- c) Translate $\hat{\beta}_{wage}$ into an approximate percent wage effect (hint: $100 \times \hat{\beta}_{wage}$ is a good approximation for small values).

6. Falsification / placebo

- a) Implement a placebo test by restricting the sample to older cohorts (e.g., *birth_cohort* $\leq$ 1960) and re-estimate the schooling model.
- b) Report and interpret the results.
- c) What would you expect to find if the design is valid?

7. Policy interpretation and limitations

- a) Provide a brief policy-facing interpretation (4–6 sentences) of your preferred estimate.
- b) List one threat to identification.
- c) List one external validity concern (generalizability).

Appendix

Data

File: indonesia_school_construction_teaching.csv

Unit of Observation: District $\times$ birth cohort

Variable	Description
<i>district</i>	district identifier (cluster unit)
<i>birth_cohort</i>	birth year
<i>exposed</i>	1 if cohort is young enough to be affected by the construction program ($birth_cohort \geq 1962$)
<i>school_intensity</i>	district-level school construction intensity (schools per 1,000 children)
<i>treat_intensity</i>	$exposed \times school_intensity$ (main treatment regressor)
<i>years_schooling</i>	completed years of schooling
<i>ln_wage</i>	log adult wage
<i>baseline_dev</i>	district <i>school_intensity</i> at baseline
<i>urban_share</i>	share of urban population in district

References

Duflo, E. (2001). Schooling and Labor Market Consequences of School Construction in Indonesia: Evidence from an Unusual Policy Experiment. *American Economic Review*, 91(4), 795–813.